

Adding, Subtracting, and Missing Numbers to 20

Answer Key

Kindergarten–Grade 1 Operations and Algebraic Thinking

Quick Answers

1. 13	4. 16	7. 14	10. 13, 4, 9
2. 7	5. 9	8. 8	
3. 8	6. 13	9. 7	

Curriculum

Level: Kindergarten–Grade 1 Operations and Algebraic Thinking

1.OA.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 12 tagged checks.

1. $8 + 5 = ?$

Step 1. A plus sign, and the blank is at the end — this one is a straight addition.

Step 2. Make a ten: 8 needs 2 more to reach 10, and 5 splits into 2 + 3. So $8 + 2 = 10$, then $10 + 3 = 13$.

Step 3. On the number line, start at 8 and hop forward 5.

13

2. $12 - 5 = ?$

Step 1. A minus sign with the blank at the end — take away.

Step 2. Get down to ten first: $12 - 2 = 10$, and there are 3 left to take away, so $10 - 3 = 7$.

Step 3. Check by adding back: $7 + 5 = 12$ ✓

7

3. $7 + \underline{\quad} = 15$

Step 1. The blank is in the middle, not at the end. The question is not “what is 7 plus something” — it is “how far is it from 7 up to 15?”

Step 2. Count on from 7: to 10 is 3 hops, and from 10 to 15 is 5 more, so $3 + 5 = 8$ hops altogether.

Step 3. Check: $7 + 8 = 15$ ✓

8

4. $9 + 7 = ?$

Step 1. Plus sign, blank at the end — addition again.

Step 2. Make a ten: 9 needs 1, and 7 splits into 1 + 6. So $9 + 1 = 10$, then $10 + 6 = 16$.

Step 3. A common near-miss here is 15 (counting one of the numbers twice). Counting on from the larger number, 9, is the reliable way.

16

5. $16 - 7 = ?$

Step 1. Minus sign, blank at the end — take away. Notice this is the undoing of problem 4.

Step 2. Think addition: $7 + ? = 16$. From 7 to 10 is 3, and from 10 to 16 is 6, so the answer is 9.

Step 3. Check: $9 + 7 = 16$ ✓

9

6. $\underline{\quad} - 4 = 9$

Step 1. This time the blank is the number we *started* with, so subtracting the two printed numbers would be wrong. We know what is left after taking 4 away.

Step 2. Put the 4 back: $9 + 4 = 13$.

Step 3. Check by doing the subtraction the problem shows: $13 - 4 = 9$ ✓

Watch for: a child who writes 5 here has computed $9 - 4$. Ask them to read the sentence aloud — “something take away 4 leaves 9” — and the error usually corrects itself.

13

7. Maya’s stickers.

Step 1. She starts with some and *gets more*, so the two groups join: this is addition.

Step 2. $6 + 8$. Make a ten: 8 needs 2, and 6 splits into $2 + 4$, so $8 + 2 = 10$ and $10 + 4 = 14$.

Step 3. Answer the question in words: Maya has 14 stickers.

14

8. Apples in the bowl.

Step 1. The family *eats* some, so apples leave the bowl: this is subtraction.

Step 2. $15 - 7$. Down to ten first: $15 - 5 = 10$, then $10 - 2 = 8$.

Step 3. Check by adding back: $8 + 7 = 15$ ✓ There are 8 apples left.

8

9. Sam’s marbles.

Step 1. Sam is not putting groups together and he is not taking any away — he wants to know how many are *missing*. The sentence is $5 + ? = 12$.

Step 2. Count on from 5: to 10 is 5 hops, then 10 to 12 is 2 more, so $5 + 2 = 7$ hops.

Step 3. Check: $5 + 7 = 12$ ✓ Sam needs 7 more marbles.

Watch for: 17 is the tell-tale wrong answer — it means the child added the two numbers instead of finding the difference.

7

10. Put it all together: the family of 9, 4 and 13.

Step 1. These three numbers make one family: the two parts are 9 and 4, and the whole is 13. Every fact in the family uses all three.

Step 2. Turn the addition around — adding in either order gives the same total, so $4 + 9 = 13$.

Step 3. Each subtraction fact takes one part away from the whole: $13 - 9 = 4$ and $13 - 4 = 9$.

Manual note: accept the two additions in either order. If a child writes $13 - 9 = 5$, have them check with the number line: hopping back 9 from 13 lands on 4.

13 4 9